Los Alamos NATIONAL LABORATORY

# Hands-on Tutorial: Parsimonious Fine-Tuning of Scientific Foundation Models That Predict Time Series Evolution of Partial Differential Equations

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# Outline of Tutorial

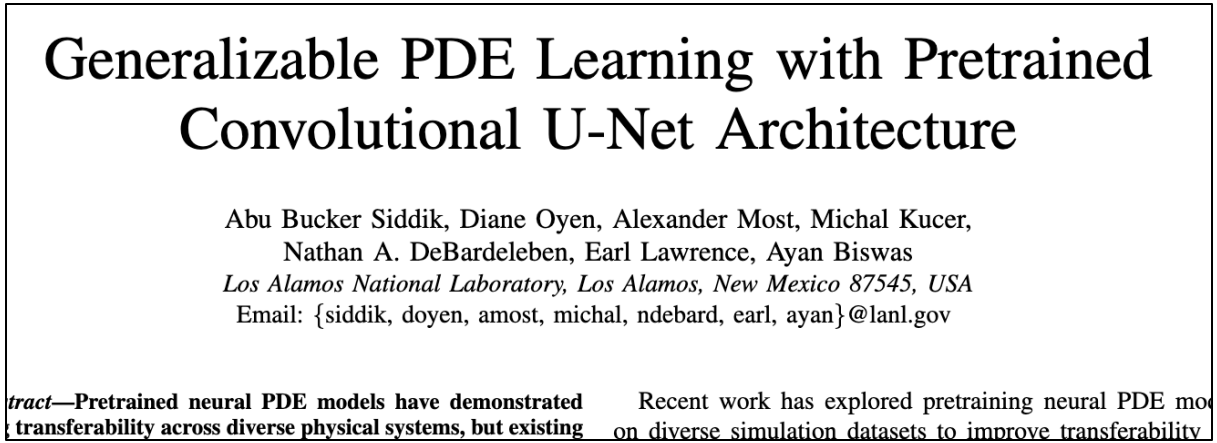
This tutorial will be 45 minutes with the following format:

(This morning before the tutorial: Configure and start VM cloud instance, which will load and assemble the data.  Will take an hour.  Bharat has coupons for NVIDIA Brev cloud.)

- (5 min) Short background presentation
- (10 min) Go to Brev instance you started earlier and start Jupyter notebook
- (15 min) Fine-tune one pretrained model together
- (15 min) Fine-tune the pretrained model with different sizes of fine-tuning datasets and make comparison output plot on a test dataset

# We Use an Efficient Foundation Model for PDEs

- Foundation models for PDEs were introduced in Nikhil's introductory talk
- The prediction task for this tutorial is to autoregressively predict 20 timesteps of a PDE trajectory unknown to the model.
- SPUS models: Small PDE U-Net Solver (https://huggingface.co/siddik-lanl)
- Already parsimonious!
- Two authors here to help with the tutorial!
- Abu Siddik and Michal Kucer



Generalizable PDE Learning with Pretrained Convolutional U-Net Architecture

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***tract*—Pretrained neural PDE models have demonstrated transferability across diverse physical systems, but existing**

Recent work has explored pretraining neural PDE mo on diverse simulation datasets to improve transferability

25th Intl. Conf. on Machine Learning and Applications (ICMLA 2026)

# Pretraining and Fine-tuning Support Parsimonious Use of Resources and Produce Quality Results

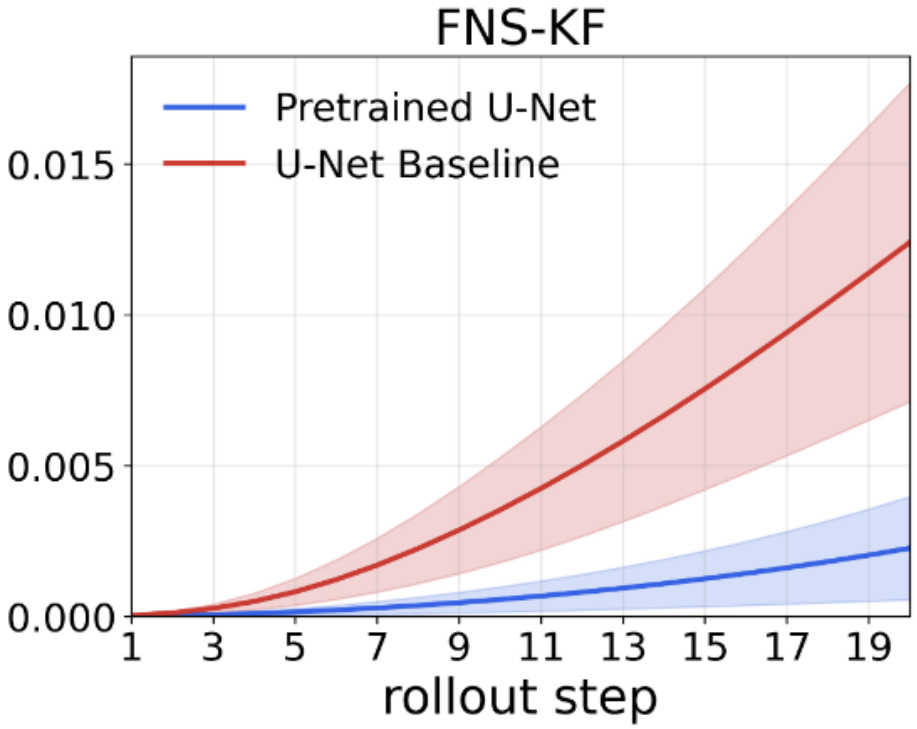
- For foundational PDE models, pretraining can be done on a variety of PDEs
  - For this tutorial, as in the paper, on four compressible Euler (CE) datasets and two Navier Stokes (NS)
- Finetuning is done on the PDE of interest
  - For this tutorial, as in the paper, Forced Navier-Stokes Kolmogorov Flow (FNS-KF)
- The pretrained model is re-used and fine-tuned with a small amount of data of interest. This reduced the data and resources required.
- For this tutorial, fine-tuning on FNS-KF used less than 1% of the amount of data used to train the full pretrained model.
- The fine-tuned model often yields better quality predictions than a model trained from scratch on the same amount of data.

# Resources Used in Various SPUS Models

| Model | Model Size | Data | Total Data | Data Units | Approx Train Time | Time Units | Training hardware | # GPUS |
|---|---|---|---|---|---|---|---|---|
| Pretrained on ~30K traj | 36m | CE+NS | 377.42 | GB | ~200 | hours | misc | 1 |
| Pretrained on ~30K traj | 9m | CE+NS | 377.42 | GB | ~90 | hours | misc | 1 |
| Pretrained on 3.6K traj | 36m | CE+NS | 50.65 | GB | ~60 | hours | misc | 1 |
| | | | | | | | | |
| Fine-tuned on 32 traj | 36m or 9m | FNS-KF | 0.42 | GB | ~1 | hours | H100 | 1 |
| Fine-tuned on 128 traj | 36m or 9m | FNS-KF | 0.68 | GB | ~4 | hours | H100 | 1 |
| Fine-tuned on 256 traj | 36m or 9m | FNS-KF | 1.03 | GB | ~8 | hours | H100 | 1 |
| | | | | | | | | |
| From Scratch on 128 traj | 36m | FNS-KF | 0.68 | GB | ~4 | hours | misc | 1 |

# Fine-tuning Produces Quality Results

- MSE of pretrained U-Net fine-tuned on FNS-KF versus U-Net trained-from-scratch on FNS-KF.
- Image courtesy Siddik, et al. ICMLA 2026.



# Participants Will Learn

- How to fine-tune a pretrained model
- How to measure resource usage and quality
- Different experiments they can perform to check quality vs resources used.
- How to use NVIDIA Brev cloud resources

# Brev (web) Setup Steps (needs to be done in advance)

- Get account on NVIDIA Brev cloud resources.  https://brev.nvidia.com
- Put in your coupon (click on "Billing", then "Redeem Code")
  - Bharat has coupons
- Follow this link to a "launchable" VM instance for the tutorial.
  - `https://brev.nvidia.com/launchable/deploy?launchableID=env-3LkozhqPdHEoKOoOzSQa8RXLeMb`
- Click on "Deploy Launchable"
- Let it start up and will take 15 minutes to build, copy data and models and assemble data.
- Don't forget to stop your instance if you are not using it, or it will keep charging at a high rate.

**Thank you to NVIDIA for the Brev coupons!**

# Brev Command Line Interface (CLI) Steps (needs to be done in advance)

```
curl -fsSL https://raw.githubusercontent.com/brevdev/brev-cli/main/bin/install-latest.sh | sed 's/release_tag="$(get_latest_release_tag)"/release_tag="v0.6.330"/' | bash

export PATH="$HOME/.local/bin:$PATH"

#Log in to Brev, brings up a web page that says you're logged in.

brev login

#show running launchables

brev ls

#create a ~/.brev/ssh_config if you have a launchable running

brev refresh

#if want to log into the launchable via the CLI, in a different terminal

ssh -F ~/.brev/ssh_config <launchable>
```

# Jupyter Notebook Steps

#check nothing is listening to port 8888

lsof -nP -iTCP:8888 -sTCP:LISTEN

#tunnel into your brev instance

ssh -F ~/.brev/ssh_config -N -L 8888:localhost:8888 <launchable>

#from brev web page compute->launchable_> start notebook

#it may not be ready so need to wait

Select the notebook  workspace/ft_tut/Notebooks/fine_tune_tut.ipynb

Select the spus_ft kernel

# Possible Experiments

## Vary FT data amount

- We will do this today

## FT on various size pretrained models

- FT with different size datasets on the 36m vs 9m pretrained models available on HF.

## FT on models pretrained differently

- HF has some models pretrained with different amounts of data. FT on those and compare

## Compare to from-scratch baseline

- HF has a FNS-KF model trained from scratch on 128 trajectories.

## Compare your own pretrained models

- Try pretraining on different PDEGym datasets

## Fine-tune on different platforms

- Make your own launchable, choose h/w, copy the tutorial startup script

Note that more models are available at https://huggingface.co/siddik-lanl